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ITAI 1378

AI Usage Documentation

Question: How do I import my dataset from Kaggle to Google Colab?

What I learned: How to obtain the API token from the Kaggle website and how to connect it with Google Colab, as well as how to download the dataset.

Question: Can you please provide me with the link to the Ultralytics documentation about how to feed the model with data, or any resource that talks about this?

What AI provided me: <https://docs.ultralytics.com/datasets/classify/#how-can-i-train-a-model-using-ultralytics-yolo> and <https://docs.ultralytics.com/modes/train/>.

Direct links to official Ultralytics' documentation on how to train a YOLO model.

Question: How do I read the training output?

```
Epoch   GPU_mem   loss  Instances   Size
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ----- 755
1/20      0G      0.4979      15      224: 100% ----- 118/118 3.2s/it 6:22
  classes  top1_acc  top5_acc 100% ----- 13/13 6.2s/it 1:20
    all      0.878      1
Epoch   GPU_mem   loss  Instances   Size
2/20      0G      0.2926      15      224: 100% ----- 118/118 3.1s/it 6:04
  classes  top1_acc  top5_acc 100% ----- 13/13 6.3s/it 1:22
    all      0.886      1
```

What I learned: I learned how to interpret the output of YOLO's training process. For example, I learned that top1_acc measures how often the model's top prediction is correct. And that top5acc measures if the correct answer is between the top 5 predictions. In my case, I realized that the top5_acc was not applicable to my project, since I only had 2 classes.

Question: Explain the structure and content of the files generated by YOLO's training.

What I learned: I learned how the files generated by YOLO's training are structured, their content, and the importance of each file. I learned that the most important files were best.pt and last.pt. The first one is used to store the weights of the best model after training. Last.pt is

used to store the weights of the last model trained, in case anything stops the training you can use last.pt to restart the training.

Question: How can I know if there is overfitting while training?

What I learned: How to properly read and interpret the results.csv file to determine if any overfitting is affecting my model, by comparing the training loss and validation loss (that both should be decreasing).

Question: Google Colab stopped my environment, how can I prevent this?

What AI provided me: It provided me with multiple solutions, such as upgrading my account to get more computational power and longer environment run times, connecting Google Drive to my project, or creating a script in the console to simulate a click every 60 seconds (so Google Colab knows I am there). I decided to connect Google Drive to my environment.

Question: I have a logic error in this function: I am getting more outputs than expected. Help me figure out what the problem is.

What I learned: The problem was that the frame count was inside the loop, so every time the loop started the frame count reset to 0.

```
def predict_video(video_path):
    cap = cv2.VideoCapture(video_path)
    frame_count = 0
    fps = cap.get(cv2.CAP_PROP_FPS)

    while cap.isOpened():
        ret, frame = cap.read()
        if not ret:
            break

        fps = cap.get(cv2.CAP_PROP_FPS)

        if frame_count % int(fps) == 0:
            results = model.predict(frame)
            class_list = results[0].probs.top1
            confidence = results[0].probs.top1conf

            if class_names[class_list] == 'fire':
                if confidence < 0.5:
                    print('Fire detected but im not sure')
                elif 0.5 <= confidence < 0.8:
                    print('Very sure it is fire')
                else:
                    print('Sure it is fire')

            frame_count += 1

    cap.release()
```

(Function after solving the logic error)

Question: I have this error trying to connect my code to Telegram. What is happening and how can I fix it? Do you have any link that directs me to the official documentation or a tutorial?

Error:

```
/tmp/ipykernel_57007/3194699934.py:7: RuntimeWarning: coroutine
'TelegramBaseClient.connect' was never awaited
```

```
client.connect()
```

```
RuntimeWarning: Enable tracemalloc to get the object allocation traceback
```

```
/tmp/ipykernel_57007/3194699934.py:9: RuntimeWarning: coroutine
'UserMethods.is_user_authorized' was never awaited
```

```
if not client.is_user_authorized():
```

```
RuntimeWarning: Enable tracemalloc to get the object allocation traceback
```

```
/tmp/ipykernel_57007/3194699934.py:23: RuntimeWarning: coroutine
'MessageMethods.send_message' was never awaited
```

```
client.send_message(receiver, message, parse_mode='html')
```

RuntimeWarning: Enable tracemalloc to get the object allocation traceback

```
<Future pending cb=[shield.._outer_done_callback()] at  
/usr/lib/python3.12/asyncio/tasks.py:922]>
```

What AI provided me: <https://core.telegram.org/bots/api>. A direct link to the Telegram API, this website had everything I needed.

Question: What is failing here?

```
request =  
requests.post(f'https://api.telegram.org/botToken/sendMessage\(chat\_id={chat\_id},text={text}\)')
```

What I learned: I learned that the arguments for `sendMessage` should be passed within the body of the POST function and not included in the URL.

Question: How do I get the ID of my Telegram bot?

What I learned: I needed to use the `getUpdates` method from the Telegram API to see some of the bot properties. I pasted the API link with the method; this method displayed a Json file for each message that the bot received.

Prompt: Generate the messages that a Fire detection system would send based on these thresholds.

```
if class_names[class_list] == 'fire':  
    if confidence < 0.5:  
        print('Fire detected but im not sure')  
    elif 0.5 <= confidence < 0.8:  
        print('Very sure it is fire')  
    else:  
        print('Sure it is fire')
```

Output:

- URGENT: Manual Verification Required - Potential Fire Activity Detected\nOur automated fire monitoring system has identified suspicious activity. The current detection confidence level is below 50%. Per safety protocols, this event requires immediate human verification to determine the nature of the activity.
- CRITICAL PRIORITY: High-Confidence Fire Detection - Emergency Protocols Required\nThe fire detection system has identified a high-probability fire event. This event has reached a critical threshold. Immediate human verification is mandatory. If the fire is visible or confirmed via remote monitoring, please contact emergency services and local authorities immediately.
- EMERGENCY: Fire Confirmed - Automated Dispatch Initiated\nCRITICAL INCIDENT ALERT - LEVEL: MAXIMUM\nEmergency services and local fire authorities have been automatically notified and are currently en route to the facility.\nIMMEDIATE INSTRUCTIONS:\n-Evacuate all personnel from the affected zone immediately.\n-Prepare to meet emergency responders at the designated arrival point.